

SELECTION IS NOT EVALUATION: INDEPENDENT CONTINUATIONS FOR RUNTIME INTER- VENTION IN LANGUAGE AGENTS

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ABSTRACT

Selecting a runtime intervention and evaluating it on the same stochastic continuation credits favorable branch noise. We test this problem with four actions, including no intervention, and two independent continuations from each shared prefix. The confirmation covers 384 identity-disjoint ALFWorld tasks (96 floorplan groups), with predictions frozen before arm outcomes. The same-draw maximum overstates cross-draw value: the planned-task gap is 15.36 percentage points (group-bootstrap 95% interval [12.46, 18.30], Holm-adjusted $p = < 10^{-4}$). Holding features, task-group folds, forest capacity, and action support fixed, direct signed-benefit prediction changes success by 1.56 points versus continuation and -0.39 points versus the strongest runnable matched controller. Neither controller contrast survives multiplicity correction. We retain the complete ScienceWorld boundary panel and adverse historical results, and make no cross-harness state-of-the-art claim. The resulting protocol turns an outcome-selected rescue into an auditable policy-value estimate.

1 INTRODUCTION

A language agent can be warned, asked to replan, or rolled back when a trajectory appears to be failing. None of these repairs is free. A warning can displace useful context, a new plan can abandon partial progress, and a rollback can undo a correct action. A runtime controller must therefore answer a treatment-choice question: which action, including no intervention, has the highest expected utility from the current prefix?

The evaluation is easy to bias. Suppose an experiment restores one agent prefix, samples several intervention branches, and chooses the branch with the best observed outcome. Scoring that choice on the same sampled branches measures a realized maximum. It credits the selection rule for favorable continuation noise. The problem is acute in binary-success environments, where one lucky branch can turn an otherwise tied action menu into an apparent rescue. Repeating the task under a different seed is not enough if the repeated outcome is again used both to choose and to score the action.

This distinction is separate from two established ideas. First, outcome regression and direct treatment-effect regression are alternative target parameterizations for heterogeneous treatment effects (Künzel et al., 2019). Second, separating selection from evaluation is a general remedy for maximization bias and adaptive reuse (van Hasselt, 2010; Dwork et al., 2015). We apply both distinctions to intervention policies for language agents. The closest same-prefix study, Zhang et al. (2026), evaluates the main controller with one realized branch per action and repeats three actions twice on 24 shallow ALFWorld prefixes as a stability check. Jiang et al. (2026) compare concrete candidate actions with a learned advisor, but their actor history, action support, intervention frequency, and compute differ from the fixed repair menu studied here.

We use two independently seeded continuations for each of four actions from a shared prefix. Any policy fitted on separate development tasks is evaluated on both draws. For an outcome-informed oracle diagnostic, the action selected on one draw is evaluated on the other. This yields a symmetric, pathwise nonnegative gap between the observed maximum and the independent-draw value of the selected action. The gap measures how much a same-outcome analysis can overstate deployable utility on the sampled task frame. It does not establish that any learned intervention helps.

The controller comparison holds the development records, observable prefix features, four task-group folds, 200-tree capacity, margin grid, and action menu fixed. One forest predicts four arm outcomes. A second predicts three signed benefits relative to continuation,

$$\Delta_a(x) = \mathbb{E}[Y_a - Y_0 \mid X = x], \quad a \in \{1, 2, 3\}. \quad (1)$$

Both controllers can choose CONTINUE. The candidate set also contains the development-best fixed arm and the strongest runnable policy from the existing arm-outcome family. All predictions were committed before confirmation arm outcomes were generated.

The confirmation panel contains 384 ALFWorld tasks balanced over six task types and grouped by 96 floorplans. Their goal directories are absent from the supplied prior configuration and outcome-exposure packet. A secondary panel contains 57 previously reserved ScienceWorld official-test variations that were not present in that packet. ALFWorld is the target population because development evidence selected it; ScienceWorld is a boundary analysis, not a second claim of broad generality. We report the complete primary family, every frozen policy, costs, early-terminal tasks, and the adverse historical evidence.

The paper makes three contributions. It gives a repeated-branch protocol and a simple diagnostic that prevents a realized action maximum from being reported as policy value. It provides a controlled target-parameterization comparison between direct signed-benefit and arm-outcome forests, with explicit no-intervention and fixed-arm fallbacks. It also releases a pre-outcome configuration, prediction freeze, bounded compute ledger, full result census, and scripts that regenerate every table and figure. The empirical conclusions are stated in Section 5; they are not widened beyond the frozen tests.

2 EVALUATING A RUNTIME INTERVENTION

2.1 POLICY VALUE AFTER A SHARED PREFIX

Let X_i be the observable prefix for task i at a fixed checkpoint. The action set is $\mathcal{A} = \{0, 1, 2, 3\}$, where 0 means continue without an injected message. A stochastic continuation under action a produces binary task success $Y_{ia} \in \{0, 1\}$ with conditional mean $\mu_a(x) = \mathbb{E}[Y_{ia} \mid X_i = x]$. For a prefix-only policy $\pi(a \mid x)$, the target value is

$$V(\pi) = \mathbb{E}_X \left[\sum_{a \in \mathcal{A}} \pi(a \mid X) \mu_a(X) \right]. \quad (2)$$

The expectation includes continuation randomness. A controller that always intervenes can have lower value than continuation even when it correctly detects a difficult prefix. Detection quality and treatment choice are different estimands (Vasudev et al., 2026).

Our simulated environments permit a repeated-branch design. From the same serialized state, we generate two draws $Y_{ia}^{(0)}$ and $Y_{ia}^{(1)}$ for every action. The request seeds are independent deterministic hashes of task, draw, action, and request index. Full-state hashes validate restoration before the first suffix call. The actor sees only the common public prefix, the assigned message, and subsequent observations.

2.2 THE SAME-DRAW OPTIMISM DIAGNOSTIC

Define $\hat{a}_i^{(r)} = \arg \max_a Y_{ia}^{(r)}$ with continuation-first tie breaking. A same-draw analysis scores this outcome-informed choice as $\max_a Y_{ia}^{(r)}$. Cross-draw evaluation instead scores the selected action on the other draw. We use the symmetric gap

$$G_i = \frac{1}{2} \left[\max_a Y_{ia}^{(0)} - Y_{i, \hat{a}_i^{(1)}}^{(0)} + \max_a Y_{ia}^{(1)} - Y_{i, \hat{a}_i^{(0)}}^{(1)} \right]. \quad (3)$$

Proposition 1. $G_i \geq 0$ for every realized pair of action-outcome vectors. If the two draws are identically distributed conditional on X_i , then $\mathbb{E}[G_i \mid X_i]$ equals the expected same-draw maximum minus the independent-draw value of the action chosen on that draw.

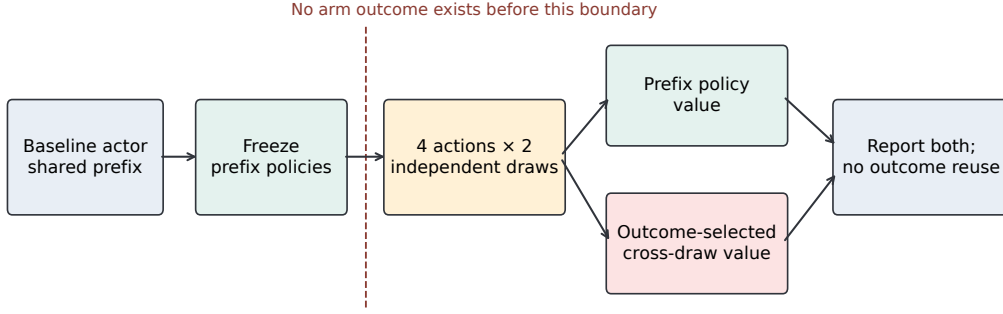


Figure 1: Repeated-branch confirmation. A baseline actor creates a checkpoint. Every eligible prefix is restored for four actions and two independently seeded draws. Prefix-only controller predictions are frozen before any of these arm outcomes. Outcome-informed action choice on draw 0 is scored on draw 1 and vice versa.

The pathwise statement follows because each maximum is at least the outcome of the action selected on the other draw. Conditional exchangeability supplies the expectation identity. Independence is needed to interpret the cross-draw score as fresh policy evaluation, but not for nonnegativity. A positive G_i can arise from unstable binary outcomes even when all actions have the same conditional mean. We therefore treat G as an evaluation diagnostic, not as treatment benefit, action heterogeneity, or controller regret. Appendix A gives the derivation and tie cases.

Figure 1 shows the information boundary. Frozen policies use the prefix only. The outcome-informed selector exists solely to quantify same-draw optimism; it is not a deployable method or a training label for the confirmation policies.

3 CONTROLLERS WITH A CONTINUATION OPTION

3.1 MATCHED TARGET PARAMETERIZATIONS

The repeated arm-outcome learner fits a multi-output random forest (Breiman, 2001) to

$$\bar{Y}_{ia} = \frac{Y_{ia}^{(0)} + Y_{ia}^{(1)}}{2}, \quad a \in \mathcal{A}. \quad (4)$$

At prediction time it compares four estimated outcome means. The direct learner uses the same records but fits three outputs,

$$\bar{D}_{ia} = \frac{1}{2} \sum_{r=0}^1 \left(Y_{ia}^{(r)} - Y_{i0}^{(r)} \right), \quad a \in \{1, 2, 3\}. \quad (5)$$

Its continuation score is fixed at zero. Thus the two learners differ in the response target while retaining the same action menu and predictor class. The comparison does not assume that direct effect regression is uniformly preferable; such an ordering is absent even in standard heterogeneous-effect settings (Künzel et al., 2019).

Both learners receive public information available at the fork: environment identifier, checkpoint index, counts of invalid or inadmissible actions, reward and usage summaries, action repetition signals, recent observations and actions, and recent prefix text. Hidden state hashes and reversibility metadata are excluded. Numeric fields are vectorized directly. Recent text uses a 512-term TF-IDF vocabulary fitted within each training fold and at most 16 truncated-SVD components.

Model selection uses four shuffled whole-task folds with seed 260908. Each candidate has 200 trees, minimum leaf size 5 or 10, and a continuation margin in $\{0, 0.05, 0.10\}$. Vocabulary and projection fitting occur inside each fold. Out-of-fold utility selects the configuration, and lower intervention rate breaks utility ties. The final forests are refit on all admitted development tasks. Source and target hashes, fold identities, feature schemas, library versions, and every candidate score are stored in the model manifests.

Table 1: Controlled comparison and frozen action menu

Component	Arm-outcome forest	Direct signed-benefit forest
Development rows	Same two-draw task blocks	Same two-draw task blocks
Features and text	Shared prefix transform	Shared prefix transform
Folds and capacity	4 groups; 200 trees	4 groups; 200 trees
Leaf and margin grid	$\{5, 10\}; \{0, .05, .10\}$	$\{5, 10\}; \{0, .05, .10\}$
Response	$(Y_0, \dots, Y_3)$	$(\bar{D}_1, \bar{D}_2, \bar{D}_3)$
No-intervention action	$a = 0$	zero-benefit reference
$a = 0$	Continue with no injected message	
$a = 1$	Check goal, observation, and previous action	
$a = 2$	Construct and execute a fresh concise plan	
$a = 3$	Undo the previous action, then replan; matched decision budget	

3.2 FALLBACKS AND THE COMPARATOR

The policy set contains five distinct decisions at confirmation time. CONTINUE always chooses $a = 0$. BEST FIXED chooses the development-best arm, which is replanning ($a = 2$) in both environments. DIRECT uses the signed-benefit learner. ARM OUTCOME uses the two-draw outcome forest and isolates response parameterization. MATCHED COMPARATOR is the strongest runnable member of the existing arm-outcome family by development out-of-fold utility. It is SINGLE-1, an otherwise matched forest fitted to one predesignated outcome draw.

The operational candidate set is $\{\text{CONTINUE}, \text{BEST FIXED}, \text{DIRECT}, \text{MATCHED COMPARATOR}\}$. Its development-only selection uses the same utility and tie rule. The selected policy is reported as SAFE SELECTED even when it is a baseline or an existing learner. In both environments the frozen selection chose SINGLE-1. Calling this rule safe means only that it can decline intervention and was selected without confirmation outcomes; it is not a safety guarantee.

4 INDEPENDENT CONFIRMATION DESIGN

4.1 ACTOR AND INTERVENTIONS

The actor is Qwen3-8B at revision b968826d9c46dd6066d109eabc6255188de91218 (Yang et al., 2025). It receives a grounded ReAct-style prompt (Yao et al., 2023): one short thought followed by an exact valid action. Thinking mode is disabled. Generation uses temperature 0.7, top- p 0.9, at most 256 output tokens, and a 32,768-token context. Serving uses deterministic-inference mode in a content-addressed SGLang container. This setting fixes the sampling implementation for the study; seeded language-model generation is not claimed to be universally invariant across hardware or serving systems (Bjarnason et al., 2026).

ALFWorld episodes have a 50-action limit and checkpoints at step 4 or 8 (Shridhar et al., 2021). ScienceWorld episodes have a 100-action limit and checkpoints at step 8 or 16 (Wang et al., 2022). A hash of each task identity assigns the checkpoint before generation. Action 3 restores the state before the last executed action and reduces the suffix limit by one. The other arms restore the checkpoint state. This gives every arm the same total decision budget. The intervention message is inserted once at the restored step and remains in history; no controller sees future state or outcome fields.

Each task first receives one baseline episode. A task that terminates before its assigned checkpoint has no intervention decision. Its observed terminal success is assigned to every policy and all contrasts and optimism terms are zero. For every eligible task, two fresh continuation arms are generated rather than copying the suffix of the baseline episode. This matters because the baseline trajectory helped define the prefix.

4.2 PANELS AND EXPOSURE BOUNDARY

The primary panel has 384 ALFWorld tasks, 64 from each of the benchmark’s six task types. Tasks come from the ALFWorld training source because it provides enough identity-disjoint instances

for a powered confirmation. Selection excluded 394 goal directories appearing in 59 supplied prior configurations or outcome records. The remaining tasks were ranked by a fixed hash seed, then assigned across eight shards. They form 96 floorplan groups. This is a fresh study panel relative to the supplied research archive, not a claim that the underlying benchmark instances have never appeared in model pretraining or outside experiments.

Development results favored ALFWorld, so the domain was chosen before the panel was sampled and before controller predictions were frozen. Our confirmatory population is therefore ALFWorld under this actor and action menu. The secondary panel has 57 ScienceWorld official-test variations drawn from a previously reserved frame after excluding 227 variations present in the supplied exposure packet. It spans 25 task families. No ScienceWorld outcome changes the primary family or controller tuning.

The panel configuration and hypotheses were committed before baseline generation. After all baselines, the feature transforms and one-hot policy choices for 406 eligible prefixes were hashed and committed before any arm outcome. The freeze contains 355 ALFWorld and 51 ScienceWorld predictions. Appendix D gives the commit and SHA-256 lineage.

4.3 HYPOTHESES, UNCERTAINTY, AND STOPPING

The ALFWorld primary family contains three quantities: mean same-draw optimism $\mathbb{E}[G]$, DIRECT minus CONTINUE, and DIRECT minus MATCHED COMPARATOR. For each, we average the paired task contrast over all 384 planned tasks. Early-terminal tasks contribute zero. We use a two-sided sign-flip test over floorplan-group sums with 50,000 Monte Carlo assignments and apply Holm correction across the three tests at familywise $\alpha = 0.05$. A positive estimate with an adjusted p value above 0.05 is not called an improvement.

Uncertainty plots show 10,000-resample percentile intervals at two levels. The task bootstrap resamples task blocks. The grouped bootstrap resamples complete floorplan groups or ScienceWorld task families and retains all tasks in a sampled group. These intervals are descriptive and marginal; the sign-flip and Holm procedure is the primary decision rule. The evaluator also reports direct-versus-arm-outcome, direct-versus-best-fixed, the development-selected policy, intervention rate, action counts, recoveries, disruptions, requests, tokens, and client wall time. Every ScienceWorld comparison is retained regardless of sign.

No task is removed for a poor outcome or long trajectory. Missing eligible cells block final analysis. One shard received a single administrative baseline resume after its original 1,000-second coordinator lease ended during an episode; its failed attempt remains in the usage ledger and the task identity, seed, code, and model were unchanged. The experiment stops when the frozen panel is complete. An actual-or-reserved occupancy cap of 72,000 H100 GPU-seconds covers model startup, qualification, generation, idle time, failures, and resumes; closed leases return unused reservation headroom. The completed study charged 47,876.7 H100 GPU-seconds (13.30 H100-hours).

5 RESULTS

The ALFWorld denominator contains all 384 planned tasks: 355 reached the frozen checkpoint and 29 ended earlier. Table 2 reports the full policy census. Direct signed-benefit prediction achieves 54.43% success and the matched controller achieves 54.82%. Relative to continuation, the direct effect is 1.56 percentage points (group-bootstrap 95% interval [-0.12, 3.41]; group sign-flip $p = 0.1048$; Holm-adjusted $p = 0.2096$). Relative to the matched controller it is -0.39 points ([-2.26, 1.57]; adjusted $p = 0.7927$). Neither primary controller comparison is significant. The development-selected safe policy changes success by 1.95 points relative to continuation. These tests answer the frozen comparisons; a favorable unadjusted subset is not substituted for them.

Direct selects continuation on 245/355 eligible prefixes, warning on 4, replanning on 106, and no rollback; it records 26 draw-level recoveries and 14 disruptions. Matched intervenes on 78.0% of prefixes, with 66 recoveries and 51 disruptions. Direct uses 27.4 calls, 42560 input tokens, and 63.4 client-seconds per planned task, versus 28.1, 43557, and 64.9 for continuation. These summaries are descriptive.

Table 2: Complete confirmation policy census. Success uses every planned task. Firing is over eligible prefixes; recoveries and disruptions count the two draw-level comparisons with continuation.

Domain	Policy	Success (%)	Fire (%)	Recover	Disrupt
ALFWorld	Continue	52.86	0.0	0	0
	Best fixed	55.34	100.0	74	55
	Direct	54.43	31.0	26	14
	Arm outcome	54.04	96.6	69	60
	Matched	54.82	78.0	66	51
	Safe selected	54.82	78.0	66	51
ScienceWorld	Continue	19.30	0.0	0	0
	Best fixed	19.30	100.0	5	5
	Direct	19.30	11.8	0	0
	Arm outcome	19.30	100.0	5	5
	Matched	14.04	100.0	2	8
	Safe selected	14.04	100.0	2	8

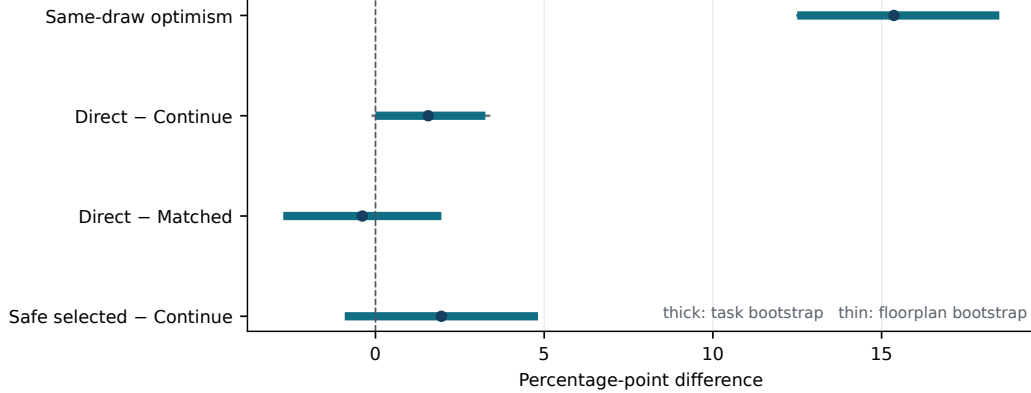


Figure 2: Confirmation effects on ALFWorld. Dots show planned-task means; thick and thin intervals show task- and floorplan-bootstrap 95% intervals. Reported adjusted p values use the prespecified group sign-flip tests and Holm correction.

Figure 2 shows both marginal task and grouped uncertainty for the primary family and the selected deployment rule. The group intervals are wider whenever shared floorplans induce material composition sensitivity. Statistical significance is determined only by the frozen group sign-flip family, not by whether a plotted interval excludes zero.

6 WHAT REPEATED BRANCHES REVEAL

The same-draw action maximum exceeds independent-draw evaluation by 15.36 percentage points over all ALFWorld tasks (task-bootstrap 95% interval [12.50, 18.49]; floorplan-bootstrap [12.46, 18.30]; Holm-adjusted $p = < 10^{-4}$). The gap is pathwise nonnegative, but its magnitude depends on how often stochastic branch outcomes disagree. It should not be read as an intervention gain or the regret of the learned controller.

Figure 3 keeps the complete group distribution visible. Groups in the upper-left quadrant exhibit a positive selection/evaluation gap while direct intervention underperforms continuation. This is precisely the case in which a realized rescue can look compelling even though the deployable policy loses utility. Groups with zero gap are retained.

ScienceWorld provides a prespecified boundary rather than a second opportunity to claim success. Its same-draw gap is 5.26 points ([1.69, 10.00] by task-family bootstrap), and Direct minus Continue is 0.00 points ([0.00, 0.00]). All 57 tasks, 25 families, six policies, costs, recoveries, and disruptions

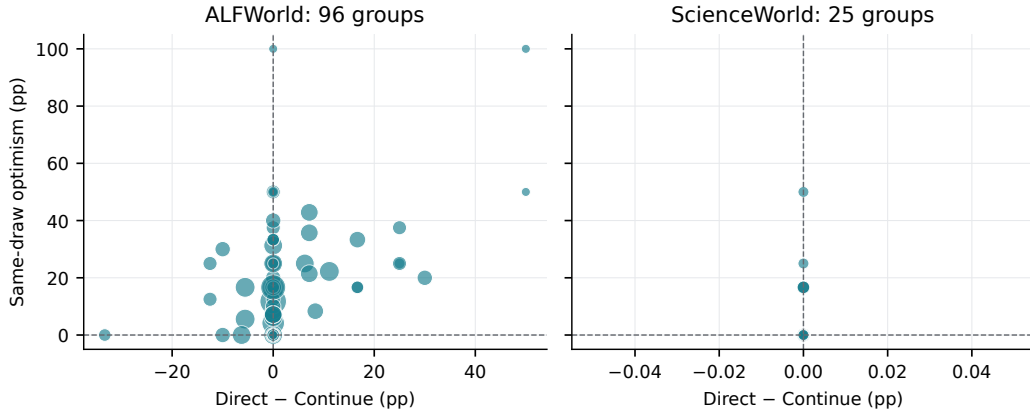


Figure 3: Group-level measurement and policy effects. Each point is a complete floorplan or task-family group; area is proportional to its task count. Vertical position is same-draw optimism and horizontal position is Direct minus Continue. Dashed lines mark zero.

appear in Table 7 and the released result JSON. Differences in direction across environments narrow the empirical scope instead of motivating post hoc domain selection.

7 CLOSEST METHODS AND COMPARISON BOUNDARY

Two current methods are closest to the decision studied here. Calibration Is Not Control fits random-forest intervention policies from same-prefix branches, but its main controller evaluation uses one realized outcome per action; repeated outcomes are reported only for three actions on 24 shallow prefixes (Zhang et al., 2026). COTA trains a 0.5B pairwise advisor to rank concrete candidate actions and reports 82.84–90.30% ALFWorld success with Qwen3-8B (Jiang et al., 2026). Its 30-step actor, environment-action candidates, repeated intervention schedule, and offline H200 budget differ from our one-shot four-message design. Neither headline score is therefore a matched estimate of our controller contrast. Our strongest matched comparator is the development-selected single-draw outcome forest run with the same actor revision, prefix history, action support, decision budget, features, and confirmation outcomes.

The technical comparison isolates two method choices. Outcome modeling estimates all four response functions; direct signed-benefit modeling estimates three effects relative to continuation, as in the response-versus-effect distinction studied by Künzel et al. (2019). Cross-draw scoring then separates maximization from evaluation, following the same statistical principle as Double Q-learning (van Hasselt, 2010). Unlike observational treatment-effect estimation, every action here is executed from a serialized simulator state. The estimand is therefore the utility of the frozen action menu, not a general causal effect or an unmatched benchmark rank.

8 LIMITATIONS, ETHICS, AND CONCLUSION

The confirmation covers one 8B actor, two text environments, one checkpoint rule, four fixed messages, binary terminal success, and two draws per action. Two draws support a paired selection/evaluation diagnostic but estimate each action’s continuation distribution coarsely. The ALFWorld tasks are new relative to the supplied archive, not guaranteed absent from model pretraining. ALFWorld was chosen using development evidence, while the smaller ScienceWorld panel is boundary evidence. Neither panel supports claims about web agents, code agents, safety-critical tools, human interventions, or arbitrary action proposals.

The direct and arm-outcome forests are controlled small-data baselines. A null direct-versus-outcome comparison may reflect target parameterization, limited features, development shift, or forest capacity. A favorable result would remain local to the frozen action menu. The group bootstrap also inherits

the benchmark’s floorplan and task-family grouping choices. More environments, actor families, intervention times, and continuation draws are required before recommending a deployment policy.

The study uses simulated household and elementary-science tasks and contains no human-subject data. Interventions can still degrade task success, so the artifact reports disruptions and includes continuation as an action. Compute is capped and fully charged by GPU occupancy rather than successful examples. The released repository remains private during double-blind review; the supplementary archive removes user names, host names, repository ownership, and absolute paths. Authors must enter affiliations and eligibility information truthfully in the submission system.

On the frozen ALFWorld panel, independent continuation evaluation establishes a 15.36-point gap relative to same-draw maximization. The matched direct controller changes success by 1.56 points versus continuation and -0.39 points versus the strongest runnable comparator. The principal result is therefore a measurement protocol and a bounded controller comparison, not a claim that intervention always helps. Runtime controllers should include no action, freeze their choices before outcomes, and value them on continuation draws not used for selection.

Reproducibility statement. Appendix H specifies task construction, prompts, seeds, software, model revision, container digest, controller grids, inference procedures, and exact commands. The anonymous supplement includes the frozen configurations and predictions, raw result hashes, failure records, evaluator, figure builder, environment lock, and a script that verifies and regenerates every reported number. The main repository commit and every derived artifact are linked by SHA-256 manifests.

AI use statement. AI tools assisted literature search through alphaXiv, experiment and analysis code, manuscript drafting, copyediting, and format checks. Procedural research skills influenced the evidence and review workflow (Kassis et al., 2026). The confirmation hypotheses, panel, inference, stopping rule, and controller predictions were frozen before their corresponding outcomes. Quantitative claims were regenerated from stored artifacts, citations were checked against source papers, and adverse results were retained. The authors remain responsible for the manuscript and submission.

REFERENCES

- Bjarni Haukur Bjarnason, André Silva, and Martin Monperrus. On randomness in agentic evals, 2026. URL <https://arxiv.org/abs/2602.07150>.
- Leo Breiman. Random forests. In *Machine Learning*, volume 45, pp. 5–32, 2001. doi: 10.1023/A:1010933404324.
- Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. Generalization in adaptive data analysis and holdout reuse. In *Advances in Neural Information Processing Systems*, volume 28, 2015. URL <https://arxiv.org/abs/1506.02629>.
- Yanze Jiang, Mingxuan Li, Yuhao Wang, Shengfang Zhai, and Jiaheng Zhang. Don’t solve, just compare: Tiny advisors for runtime intervention in LLM agents, 2026. URL <https://arxiv.org/abs/2608.21027>.
- Timothy Kassis, Vinayak Agarwal, Yuhuan He, Darshil Patel, and Aubrey M. Brueckner. Scientific agent skills: A library of procedural knowledge for research agents, 2026. URL <https://arxiv.org/abs/2609.00065>.
- Sören R. Künzle, Jasjeet S. Sekhon, Peter J. Bickel, and Bin Yu. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165, 2019. doi: 10.1073/pnas.1804597116.
- Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. ALFWorld: Aligning text and embodied environments for interactive learning. In *International Conference on Learning Representations*, 2021. URL <https://arxiv.org/abs/2010.03768>.
- Hado van Hasselt. Double Q-learning. In *Advances in Neural Information Processing Systems*, volume 23, 2010.

- Rakshith Vasudev, Melisa Russak, Dan Bikel, and Waseem Alshikh. Accurate failure prediction in agents does not imply effective failure prevention, 2026. URL <https://arxiv.org/abs/2602.03338>.
- Ruoyao Wang, Peter Jansen, Marc-Alexandre Côté, and Prithviraj Ammanabrolu. ScienceWorld: Is your agent smarter than a 5th grader? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 11279–11298, 2022. doi: 10.18653/v1/2022.emnlp-main.775. URL <https://aclanthology.org/2022.emnlp-main.775/>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report, 2025. URL <https://arxiv.org/abs/2505.09388>.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2210.03629>.
- Chubin Zhang, Zhenglin Wan, Xingrui Yu, Jingxuan Wu, Qi Wen, Pengfei Zhou, Wangbo Zhao, and Ivor Tsang. Calibration is not control: Why LLM-agent oversight needs intervention, 2026. URL <https://arxiv.org/abs/2606.21399>.

A DERIVATION OF THE REPEATED-BRANCH DIAGNOSTIC

For a fixed task, write $m^{(r)} = \max_a Y_a^{(r)}$ and $\hat{a}^{(r)} = \arg \max_a Y_a^{(r)}$, with a fixed continuation-first order for ties. For either draw,

$$m^{(0)} - Y_{\hat{a}^{(1)}}^{(0)} \geq 0, \quad m^{(1)} - Y_{\hat{a}^{(0)}}^{(1)} \geq 0, \quad (6)$$

because the selected action from the other draw remains an element of the maximization set. Their mean is G , so $G \geq 0$ pathwise. The result is unchanged by ties because any deterministic tie choice indexes an available action.

Now condition on the shared prefix $X = x$. If the two action-outcome vectors are exchangeable and independent draws from the same conditional continuation distribution, then

$$\mathbb{E}[G \mid x] = \mathbb{E}[m^{(0)} \mid x] - \mathbb{E}[Y_{\hat{a}^{(1)}}^{(0)} \mid x] \quad (7)$$

$$= \mathbb{E}\left[\max_a Y_a \mid x\right] - \mathbb{E}[\mu_{\hat{a}(Y')}(x) \mid x], \quad (8)$$

where Y' is an independent action-outcome vector and $\hat{a}(Y')$ is the action selected from it. The first term is a realized maximum. The second is the value of the outcome-informed selection rule on a fresh continuation. It is not the value of any prefix-only deployable controller. If all actions share the same success probability but outcomes fluctuate, the difference may still be positive; G therefore identifies evaluation optimism rather than a causal intervention effect.

B ACTION AND INFORMATION CONTRACTS

Table 3: The complete frozen action menu. The message is inserted once immediately before the next actor decision and remains in the visible transcript.

Arm	Name	Operation
0	CONTINUE	Restore the checkpoint and append no message.
1	WARNING	“Re-check the goal, the latest observation, and your previous action. Continue only after correcting any inconsistency you notice.”
2	REPLAN	“Construct a fresh, concise plan from the current goal and observable state. Then execute that plan.”
3	ROLLBACK-REPLAN	Restore environment and dialogue to before the latest reversible action, append “The previous attempted action has been reverted. Construct a fresh, concise plan from the current goal and restored observable state. Then execute that plan,” and reduce the suffix budget by one decision.

Every arm receives the same task instruction, public action history, public observations, actor weights, decoding parameters, and total decision budget. The controller receives only fields computed from the prefix. Environment state hashes, future observations, suffix outcomes, rollback metadata, and task-selection provenance are excluded. Arm 3 changes the simulator state by exactly the most recent reversible action and has no special access to a future legal-action sequence. The intervention decision occurs once per episode.

C CONTROLLER CONSTRUCTION AND DEVELOPMENT SELECTION

The 50 numeric features are action-class fractions; counts and rates of admissible, repeated, invalid, and failed actions; reward summaries; observation and prompt lengths; token usage; action and observation novelty; vocabulary growth; and step index. The text field concatenates the recent public observations and actions. Word TF-IDF uses at most 512 terms; the fitted development vocabulary is reduced to at most 16 components. All text fitting occurs inside each task-group fold during development selection. The final transform is refit on all admitted development tasks only.

The repeated arm-outcome control, REPEATED, predicts all four outcomes after fitting to the mean of two draws. SINGLE-1 uses the same action-outcome formulation but fits to one predesignated draw; it was the strongest runnable development comparator. DIRECT predicts three outcomes relative to

Table 4: Development-only policy selection. Utilities average the two saved continuation draws within each complete task row. The strongest matched comparator and the operational selection were fixed before confirmation outcomes.

Environment	Candidate	OOB utility (%)	Intervention rate (%)
ALFWorld	CONTINUE	45.35	0.00
	BEST FIXED (arm 2)	50.39	100.00
	DIRECT	47.67	34.11
	SINGLE-1	51.16	79.84
ScienceWorld	CONTINUE	20.91	0.00
	BEST FIXED (arm 2)	24.55	100.00
	DIRECT	23.64	21.82
	SINGLE-1	27.27	100.00

continuation. Both target-parameterization learners share the same feature builder, task-group folds, 200-tree random-forest class, minimum-leaf grid $\{5, 10\}$, continuation-margin grid $\{0, .05, .10\}$, and task rows. A candidate must exceed the continuation margin to intervene; otherwise it chooses arm 0. Utility ties favor the lower firing rate and then the prespecified candidate order.

D PANEL CONSTRUCTION AND LINEAGE

The ALFWorld source inventory was grouped by its six task types. We removed every goal directory found in the 59 supplied prior configuration and outcome files, ranked the remaining identities by SHA-256 with seed 270914, and retained the first 64 identities per type. The resulting 384 tasks occupy 96 floorplan groups. ScienceWorld began from a 57-variation reserved official-test frame. We verified that none of its variations occurred among the 227 variations in the supplied exposure packet; the panel occupies 25 task families. These exclusions establish independence from the supplied research record, not from model pretraining or all work by other parties.

Table 5: Pre-outcome lineage. Full digests are machine-checked in the supplement.

Object	Frozen identifier
Panel and analysis commit	8ddf66b and 4dd0785 precede baseline generation
Prediction-freeze commit	f0f26bf, pushed before arm generation
Eligible prefixes	406: 355 ALFWorld and 51 ScienceWorld
Prefix digest	0f5a80d1...46e863
Prediction digest	78380914...f6911b
Arm-outcome model digest	759b9d02...7d612b
Direct model digest	4ad68910...fe16db

Baseline generation produced 35 structural early terminations. One episode exceeded its original coordinator lease. A single administrative resume reused the frozen task, seed, actor, software, and checkpoint rule; the interrupted attempt and resumed occupancy are both retained in the append-only ledger. No arm outcome was available when features and actions for the remaining 406 prefixes were serialized and committed. During arm generation, eight serving endpoints became unavailable after roughly one hour, leaving 560 canonical cells with technical failure flags. The evaluator rejected that state. The administrative resume archived every failed cell as an `attempt1` record and regenerated only those cells with their original configuration, prefix, action, draw, and request seeds. A separate server preflight failed on an occupied port and generated no outcome; restarting it on a different port completed the shard. No second-attempt record exists, and no failed cell enters the analysis.

E COMPLETE STATISTICAL SPECIFICATION

For a policy p , task-level utility is the mean of the two selected arm outcomes. Structural early-terminal tasks contribute their observed terminal success to every policy. A policy contrast is the paired difference in task utility. Task-bootstrap intervals resample tasks with replacement. Group-

bootstrap intervals resample floorplans for ALFWorld and task families for ScienceWorld, retaining every task in each sampled group. Both use 10,000 draws and fixed stored seeds.

The primary randomization statistic is the absolute mean task contrast. Under the null, we independently multiply each group sum by -1 or $+1$, retaining within-group task dependence, and compare 50,000 Monte Carlo assignments with the observed statistic. The Monte Carlo p value includes the observed assignment through the usual plus-one correction. We order the three raw ALFWorld p values and apply Holm’s step-down adjustment. The family comprises the optimism gap, DIRECT minus CONTINUE, and DIRECT minus SINGLE-1. All other tests and intervals are descriptive secondary analyses.

Recoveries count draw-level cases in which continuation fails and the policy succeeds. Disruptions count the reverse. Intervention rate is computed over eligible prefixes; success denominators include all planned tasks. Request, token, and client-wall-time summaries report the selected suffix for eligible tasks. The release additionally preserves all recorded calls and the GPU lease ledger. No missing eligible cell is imputed: the final evaluator refuses to emit results if any of the eight required arm/draw cells is absent.

F COMPLETE CONFIRMATION RESULTS

Table 6: All ALFWorld contrasts. Values are percentage points. Holm adjustment applies only to the first two rows and the same-draw diagnostic in the text.

Contrast	Mean	Task 95%	Group 95%	Group p	Holm p
Direct — Continue	1.56	[0.00, 3.26]	[-0.12, 3.41]	0.1048	0.2096
Direct — Matched	-0.39	[-2.73, 1.95]	[-2.26, 1.57]	0.7927	0.7927
Direct — Arm outcome	0.39	[-2.21, 2.99]	[-1.82, 2.64]	0.8228	—
Direct — Best fixed	-0.91	[-3.39, 1.56]	[-3.09, 1.31]	0.4881	—
Safe selected — Continue	1.95	[-0.91, 4.82]	[-0.38, 4.26]	0.1311	—

Table 7: All ScienceWorld contrasts, retained as unadjusted boundary analyses. Values are percentage points.

Contrast	Mean	Task 95%	Group 95%	Group p	Holm p
Direct — Continue	0.00	[0.00, 0.00]	[0.00, 0.00]	1.0000	—
Direct — Matched	5.26	[-0.88, 12.28]	[-0.81, 11.86]	0.1860	—
Direct — Arm outcome	0.00	[-5.26, 5.26]	[-4.17, 4.39]	1.0000	—
Direct — Best fixed	0.00	[-5.26, 5.26]	[-4.17, 4.39]	1.0000	—
Safe selected — Continue	-5.26	[-12.28, 0.88]	[-11.86, 0.81]	0.1860	—

ALFWorld same-draw optimism is 15.36 points with task-bootstrap interval [12.50, 18.49], group-bootstrap interval [12.46, 18.30], raw group sign-flip $p = < 10^{-4}$, and Holm-adjusted $p = < 10^{-4}$. ScienceWorld optimism is 5.26 points with task-family-bootstrap interval [1.69, 10.00] and unadjusted group sign-flip $p = 0.0632$.

Across baseline, arm, and retained attempt records, the episode ledger contains 127,213 model requests, 253,758,979 input tokens, 6,921,686 output tokens, and 301,110.6 client-seconds. It records 561 failed requests and 561 requests with unknown token usage. The separate GPU lease ledger charges model loading, qualification probes, generation, idle occupancy, failures, and cleanup.

G HISTORICAL AND EXPLORATORY EVIDENCE

Earlier work motivated the new confirmation but does not enter its multiplicity family. On a 134-game official unseen ALFWorld panel with a different actor interface, the prior repeated-branch learner reached 68.16% against 69.40% for continuation, a difference of -1.24 percentage points with task-bootstrap interval $[-4.98, 2.24]$ and $p = 0.5946$. Same-arm mismatches ranged from 8.25% to 12.06%, and the same-draw sample oracle exceeded a fresh-replica evaluation by 7.94 points. A four-stratum retrospective direct-benefit reanalysis averaged 30.14%, compared with 32.62% for continuation and 28.92% for the repeated-outcome policy. These records motivated independent continuation evaluation and the direct target comparison; they do not confirm either new hypothesis.

Table 8: Mean selected-suffix cost per planned task. Structural early terminations have zero suffix cost.

Domain	Policy	Requests	Input tokens	Output tokens	Client seconds
ALFWorld	Continue	28.1	43557	1508	64.9
	Best fixed	27.5	43237	1468	63.2
	Direct	27.4	42560	1470	63.4
	Arm outcome	27.7	43559	1481	63.8
	Matched	27.7	43459	1487	64.1
	Safe selected	27.7	43459	1487	64.1
ScienceWorld	Continue	47.7	177369	2836	124.6
	Best fixed	48.3	193820	2837	126.5
	Direct	47.7	178377	2847	125.3
	Arm outcome	48.3	193820	2837	126.5
	Matched	52.8	199333	3139	138.0
	Safe selected	52.8	199333	3139	138.0

The native-interface follow-up and its demonstration ablation are reported only as actor-side exploratory analyses. Changing thinking mode, output cap, and demonstration use together produced 80.60% versus 70.90% in one paired 134-task draw. A later conditional ablation held thinking and cap fixed: including two released demonstrations produced 116/134 successes (86.57%) versus 107/134 (79.85%), a +6.72-point estimate with task-bootstrap interval $[-0.75, 14.18]$ and two-sided sign-flip $p = 0.1087$. Excluding one scene reversed the latter estimate to -3.51 points. Neither result is pooled with confirmation, establishes a controller effect, or supports a state-of-the-art claim.

A separate 27B experiment is retained as a systems result. On its internal reserved panel, the plain 27B actor scored 131/160 (81.88%) against 76/160 (47.50%) for an 8B self-check system; the paired difference was +34.38 points with a prespecified 97.5% cluster- t interval $[24.20, 44.55]$. The systems differ in model, policy, filtering, and decoder stream, and 49 of 64 confirmation floorplans had appeared in historical tasks. On the historically exposed 134-task public ALFWorld frame, the plain and self-check 27B systems both scored 108/134. These results establish neither an isolated parameter-count effect nor a controller advance and are not pooled with confirmation. CAMELS-CIS is outside this paper. The released evidence index preserves those boundaries rather than using favorable actor or domain subsets to replace the frozen confirmation.

H REPRODUCTION

The anonymous supplement contains the exact ICLR source, panel manifests, frozen prefix features and actions, controller manifests, raw confirmation records, analysis environment lock, tests, evaluator, rendering program, and artifact manifest. Model weights and benchmark installations are excluded. Analysis and paper regeneration require no GPU.

```

uv venv --python 3.12 .venv
uv sync --frozen
.venv/bin/python -m pytest -q
mv artifacts/confirmation artifacts/confirmation.release
.venv/bin/python controller/evaluate_confirmation.py \
  --config configs/independent-panel/panel-shard0.json \
  --config configs/independent-panel/panel-shard1.json \
  --config configs/independent-panel/panel-shard2.json \
  --config configs/independent-panel/panel-shard3.json \
  --config configs/independent-panel/panel-shard4.json \
  --config configs/independent-panel/panel-shard5.json \
  --config configs/independent-panel/panel-shard6.json \
  --config configs/independent-panel/panel-shard7.json \
  --raw raw \
  --predictions artifacts/prediction-freeze/predictions.json \
  --prediction-freeze \

```

```

artifacts/prediction-freeze/prediction-freeze.json \
--out artifacts/confirmation
.venv/bin/python paper/build_assets.py
cd paper
mkdir -p build
pdflatex -interaction=nonstopmode -halt-on-error \
-output-directory=build main.tex
BIBINPUTS=.: BSTINPUTS=.: bibtex build/main
pdflatex -interaction=nonstopmode -halt-on-error \
-output-directory=build main.tex
pdflatex -interaction=nonstopmode -halt-on-error \
-output-directory=build main.tex
cd ..
python tools/verify_artifacts.py --supplement \
../intervene-or-continue-iclr2027-supplement.zip

```

Inference reproduction additionally requires ALFWorld 0.4.2, TextWorld 1.7.0, ScienceWorld 1.3.0, the recorded Qwen3-8B revision, and the content-addressed serving image. Each GPU worker receives one panel shard and a unique port. A bounded coordinator enforces a 72,000 H100 GPU-second actual-or-reserved occupancy cap; closed leases return unused reservation headroom. The immutable ledger records 47,876.7 charged H100 GPU-seconds across 26 closed leases. Exact request seeds are derived from task identity, arm, draw, and request index. New stochastic generations are replications and must not overwrite the released outcomes.

The supplement’s verification program checks the source commit, all protected input hashes, complete task and cell counts, task/fold disjointness, action visibility, panel exclusions, prediction freeze, recomputed result JSON, figure inputs, page limit, unresolved references, anonymous strings, and final archive digest. Absolute execution paths and private repository ownership are replaced by symbolic roots in the anonymous archive; scientific identifiers and file-content digests are unchanged.

I AI ASSISTANCE AND AUTHOR RESPONSIBILITIES

AI tools were used for literature discovery through alphaXiv, code drafting, statistical and manuscript review, LaTeX drafting, copyediting, and format inspection. They were not treated as authors. The pre-outcome commits and immutable result evaluator separate assistance during design from outcome-conditioned reporting. The complete result census, adverse evidence, source links, and assistance ledger are provided for audit. Human authors remain responsible for checking claims, citations, authorship, conflicts, affiliations, eligibility, disclosure categories, and the final submission.